WEB-BASED ML MODEL EVALUATOR

System Validation Report

Project	Web-Based ML Model Evaluator (Team SE-Jackfruit)
Backlog item	BL-14 - Unit Test Report and System Validation Report
Validated against	Software Requirements Specification (documents/requirements.pdf): FR-01 to FR-08, NFR-01 to NFR-05, security requirements
Test run	4 November 2026 (automated suite 20:50 IST, manual cases the same evening)
Source of results	docs/test-results (pytest-verbose.txt, junit.xml, perf-100mb-output.txt) and the test plan

1. Purpose

Validation answers the question: *does the finished system do what the requirements say?* This report checks every functional and non-functional requirement of the SRS against the acceptance criteria written there, using the test cases of the test plan. It then states for each requirement whether it is validated.

2. Method and environment

Item	Detail
System tested	The application on the main branch after the BL-11 merge (Flask web application, SQLite storage, scikit-learn evaluation).
Automated cases	pytest tests that drive the real application over HTTP with real model and CSV files, plus service-level tests. Command: python -m pytest tests -v -p no:cacheprovider --junitxml=junit.xml
Manual cases	NF_PF_03 (timing with a model close to 100 MB) and NF_US_01 (new-user usability check).
Environment	Windows, Python 3.14.6, pytest 9.1.1, run from the backend folder on 4 November 2026 at 20:50 IST
Pass rule	A requirement is validated when every test case linked to it passes and no acceptance criterion in the SRS is left uncovered.

3. Overall result

Test cases in the plan	Passed	Failed	Not executed	Defects found
86	86	0	0	0

All 86 cases of the plan were executed and passed. The automated suite ran 219 tests with no failures. Every requirement below is validated; a few carry a limitation noted in section 5.

4. Requirement-by-requirement validation

For each requirement: what the SRS asks, the test cases that check it, and the verdict. Case IDs are explained in the test plan; each case names its automated test or manual procedure.

Req.	Requirement (summary)	Test cases	Cases	Result	Verdict
FR-01	Register and log in. An Analyst can register and log in; no one can become Admin through registration; an Admin can only be created by an existing Admin.	SEC_01, SEC_02, SEC_03, SEC_04, SEC_06, 01_UT_01, 01_UT_02, 01_UT_03, 01_UT_04, 01_ST_01, 02_UT_01, 02_UT_02, 02_IT_01, 02_IT_02, 02_ST_01, 13_ST_01	16	16 of 16 pass	**Validated**
FR-02	Upload and validate files. Dataset (with target) and models are checked individually at upload; each model is checked for compatibility with the dataset before it can be evaluated; bad files are rejected with a reason.	03_UT_01, 03_UT_02, 03_UT_03, 03_UT_04, 03_IT_01, 03_IT_02, 04_UT_01, 04_UT_02, 04_UT_03, 04_IT_01, 04_IT_02, 04_IT_03, 06_IT_02, 06_ST_01	14	14 of 14 pass	**Validated**
FR-03	Select task type. Classification or regression is chosen; the task is checked against the model and dataset before evaluation.	05_UT_01, 05_IT_01, 05_ST_01, 06_IT_02, 06_ST_01	5	5 of 5 pass	**Validated**
FR-04	Run model and compute metrics. Classification gives accuracy, precision, recall, F1, ROC-AUC; regression gives MAE, MSE, RMSE, R2; values match a known test case.	06_UT_01, 06_UT_02, 06_UT_03, 06_ST_01, 06_ST_02, 07_UT_01	6	6 of 6 pass	**Validated**
FR-05	Visualizations. Confusion matrix (and ROC when probabilities exist) for classification; residual plot for regression; charts match the results.	06_UT_03, 06_ST_01, 08_UT_01, 08_UT_02, 08_IT_01, 08_ST_01, 08_ST_02	7	7 of 7 pass	**Validated**
FR-06	Compare evaluations. Two or more completed evaluations are shown side by side; different task types are not compared without a warning.	09_UT_01, 09_IT_01, 09_IT_02, 09_ST_01	4	4 of 4 pass	**Validated**
FR-07	Saved evaluations and reports. Evaluations are saved and reopen with the same values; reports (single or multiple) are built from saved data.	06_IT_01, 07_IT_01, 10_UT_01, 10_UT_02, 11_UT_01, 11_UT_02, 11_UT_03, 11_IT_01, 11_ST_01	9	9 of 9 pass	**Validated**
FR-08	Admin: accounts and roles. Admins see all accounts, suspend and unsuspend, change roles, with an audit log; cannot suspend themselves; the last Admin is protected.	SEC_06, SEC_07, 02_IT_02, 02_IT_03, 02_IT_04, 12_UT_01, 12_UT_02, 12_IT_01, 12_IT_02, 12_ST_01, 13_UT_01, 13_IT_01, 13_IT_02, 13_IT_03, 13_ST_01	15	15 of 15 pass	**Validated**
NFR-01	Performance. Up to 50,000 rows and a 100 MB model: result within 10 s for a single user (95 % of runs).	03_UT_04, NF_PF_01, NF_PF_02, NF_PF_03	4	4 of 4 pass	**Validated**
NFR-02	Security: data isolation. One Analyst cannot open another's datasets, models or evaluations, even by guessing a URL and ID.	SEC_08, SEC_09, SEC_10, SEC_11, SEC_12	5	5 of 5 pass	**Validated**
NFR-03	Reliability: clear errors. Corrupt file, schema mismatch and oversize file each return a specific, informative message and never crash the session.	03_UT_02, 04_UT_02, 04_UT_03, 04_IT_01, 06_ST_03, 07_UT_02, 11_ST_01, 12_ST_01, NF_ER_01, NF_ER_02, NF_ER_03, NF_ER_04	12	12 of 12 pass	**Validated**
NFR-04	Usability. A new user can upload a file and reach the results unassisted.	NF_US_01, NF_US_02	2	2 of 2 pass	**Validated**
NFR-05	Extensibility. A new metric, visualization or task type can be added without changing unrelated code.	NF_EX_01	1	1 of 1 pass	**Validated**
Security	Security objectives. Access control on every page and API call; CSRF protection; passwords stored hashed; safe redirects.	SEC_01, SEC_02, SEC_03, SEC_04, SEC_05, SEC_06, SEC_07, SEC_08, SEC_09, SEC_10, SEC_11, SEC_12	12	12 of 12 pass	**Validated**

5. Evidence for the main end-to-end flows

These system-level cases walk through complete user journeys and are the strongest evidence that the core use cases work together:

Case	Journey	Result
06_ST_01	Upload dataset and model, validate, choose classification, evaluate, view metrics and charts (end to end)	Pass
06_ST_02	Same journey for regression; several models evaluated on one dataset	Pass
09_IT_02	Compare evaluations of the same task side by side	Pass
11_UT_01 / 11_UT_02	Download a PDF report for one and for several evaluations	Pass
13_UT_01 / 12_UT_02	Admin changes a role and suspends / reactivates an account; actions appear in the audit log (13_IT_03)	Pass
SEC_09 / SEC_10	A second Analyst cannot reach the first one's files, evaluations, charts or reports by ID	Pass

6. Limitations and observations

Validation is only as strong as the tests behind it. These points are stated openly so the results are not over-read:

NFR-01. Measured on one Windows laptop (single user). Both 50,000-row runs of the suite and five runs with a 95.8 MB random-forest model (1.59 to 1.67 s) were far below 10 s. Only one kind of large model was tested.

NFR-04. One classmate who was not on the team and had not seen the app finished the whole flow unaided in about 2-3 minutes with no hesitation. A single tester is a small sample.

NFR-05. Checked by adding a chart type through the chart module only. Adding a new metric or task type was not exercised in a test; the design keeps them in separate modules.

Security. Covers the security section of the SRS (access control, CSRF, password storage). No penetration testing was done beyond these cases.

FR-04. Metric values are compared with hand calculations on small fixed datasets.

Evolution since the first design. During Sprint 3 the storage moved from JSON files to SQLite and the interface became a multi-page application with login and roles. These changes support the same requirements; the test plan was updated to match (BL-12).

Known gaps outside the SRS. There is no password-reset function, and files uploaded before the move to SQLite must be uploaded again. Neither is a requirement of the SRS.

7. Conclusion

All 8 functional requirements, all 5 non-functional requirements and the security objectives are validated: 86 of 86 test cases passed and no defects were found. The system meets the acceptance criteria of the SRS, with the limitations in section 6.